

Finite-Time Synchronization of Networks via Quantized Intermittent Pinning Control

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Abstract—This technical correspondence considers finite-time synchronization of dynamical networks by designing aperiodically intermittent pinning controllers with logarithmic quantization. The control scheme can greatly reduce control cost and save both communication channels and bandwidth. By using multiple Lyapunov functions and convex combination techniques, sufficient conditions formulated by a set of linear matrix inequalities are derived to guarantee that all the node systems are synchronized with an isolated trajectory in a finite settling time. Compared with existing results, the main characteristics of this paper are twofold: 1) quantized controller is used for finite-time synchronization and 2) the designed multiple Lyapunov functions are strictly decreasing. An optimal algorithm is proposed for the estimation of settling time. Numerical simulations are provided to demonstrate the effectiveness of the theoretical analysis.

Index Terms—Finite time, intermittent pinning, quantized controller, synchronization.

I. INTRODUCTION

In real world, many natural and artificial systems, including the Internet and electrical power grids, can be presented by dynamical networks (DNs). In the past two decades, network systems have been extensively investigated across different disciplines due to that dynamical behaviors of network systems are more complex than any individual subsystems and have many important applications in automatic control and secure communication [1], [2]. Synchronization means that two or more dynamical systems adjust each other to lead to a common dynamical behavior [3], [4]. In case that the interconnections among subsystems (nodes) are not strong, a network cannot achieve synchronization if no controller is added to the nodes. Up to date, many effective control methods have been proposed for

synchronization of networks, such as pinning control [5]–[8], impulsive control [9], [10], intermittent control [11]–[13], and adaptive control [14].

Pinning control, where only a small fraction of nodes are controlled, is an effective approach to synchronize DNs [15]–[20]. However, most of existing pinning controllers are continuously dependent on the states of both controlled nodes and goal system. In practical applications, the transmitted signals are inevitably affected by external perturbations, which make the signals weak or interrupted intermittently. In this case, the continuous-time pinning control strategies are not applicable. Hence, pinning periodically intermittent control schemes were proposed [13], [21], [22]. Moreover, intermittent pinning control can greatly reduce control cost and the amount of the transmitted information. Considering the fact that the requirement of periodicity is not necessary in practice, aperiodically intermittent pinning controls were considered in [12] and [23]. Recently, another kind of aperiodically intermittent pinning control called event-triggered pinning control was also proposed [24], [25]. For event-triggered control, the controller works when triggering function reaches certain threshold. It is worth mentioning that most of existing intermittent pinning control schemes are designed by directly using the information of synchronization errors.

Due to the limited bits rate of communication channels and the limited bandwidth, communication constraints play a major role in stabilization and related problems of DNs and control theory. The way of representing such constraints is to use event-triggered controls [24], [25] or quantize the signals before transmission [26]–[28]. The authors of [27]–[29] considered average consensus of multiagent networks with uniform quantizers and logarithmic quantizers, respectively. Liu *et al.* [30] considered the stabilization for a class of linear systems with nonlinear disturbances by using quantized feedback control. In [31], stabilization for nonlinear discrete-time systems via quantized output feedback control. Liu *et al.* [32] investigated quantized tracking problem for a class of stochastic nonlinear uncertain systems. However, to the best of our knowledge, although quantized control has been extensively studied in the literature, few authors considered quantized intermittent control. Obviously, quantized intermittent control can further save channel resources and reduce the channel blocking.

It is well known that finite-time techniques have more advantages over asymptotical ones. Besides fast convergence rate, finite-time techniques have better robustness and disturbance rejection properties [33], [34]. Therefore, stabilization and synchronization of various systems in finite time have been extensively studied. Especially, finite-time synchronization and consensus of DNs attracted considerable attention. For instance, by developing some new analytical techniques, finite-time cluster synchronization for a class of DNs was studied in [19] and [33]. Finite-time synchronization of DNs with Markovian topology and impulsive effects was investigated in [34]. The finite-time synchronization for a class of coupled nonlinear systems was studied in [35] by using homogeneity technique. Finite-time consensus of coupled Euler–Lagrange systems was realized in [36] by using sliding mode control. Finite-time synchronization

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of switched coupled neural networks was considered in [37]. It is well known that state feedback controllers need to precisely and incessantly receive the signal of the state. Actually, most of existing results on finite-time stabilization were obtained under the framework of error-free transmitted information of the signals, and few authors consider the same problems by utilizing quantized controllers. From the view point of practical applications, finite-timely synchronizing DNs by quantized pinning control in an aperiodically intermittent way can greatly reduce control cost and save both communication channels and bandwidth.

It should be noted that, though different intermittent pinning control have been proposed, the general idea behind them is that designed Lyapunov functions increase exponentially when there is no control, and they decrease exponentially when controllers work. This idea is also usually applied to stabilization and synchronization of asynchronous switching systems [38]–[41], where multiple Lyapunov functions are usually designed. However, if the above mentioned idea is applied to finite-time stabilization and synchronization of switched systems, the finite-time stability theorem in [42] cannot be directly utilized since there is a jump of the multiple Lyapunov functions at switching instants. Moreover, a traditional way to cope with multiple Lyapunov functions is to assume that there is a sudden change and increasing of the value of multiple Lyapunov functions at switching instants, then τ (the value of τ is smaller than switching intervals) time units after the switching, it is less than the value it had just prior to the switching. This method induces great difficulty in estimating the settling time. Actually, the value of multiple Lyapunov functions after switching instants can be equal to, or more than, or less than the value just before the switching. Obviously, always assuming that the values of multiple Lyapunov functions after switching instants are larger than their values just before the switching may lead to conservative results.

Based on the above discussions, this paper investigates finite-time synchronization of DNs by designing a class of quantized semi-pinning controllers [33]. The signals of the controllers are quantized before transmission and the controllers active in an aperiodically intermittent way. The topology of the network can be symmetric or asymmetric. By designing multiple Lyapunov functions and using convex combination technique, sufficient conditions are derived. The main contributions are listed as follows: 1) the quantized intermittent pinning control needs very small amount of information, which makes the controllers more economic and practical than existing intermittent pinning controls; 2) to the best of the authors' knowledge, it is the first time that finite-time synchronization is considered by using quantized semi-pinning control; 3) sufficient conditions are formulated by a set of linear matrix inequalities (LMIs), which are different from most of existing algebraic inequality conditions for finite-time stabilization and synchronization; and 4) based on the convex combination technique, the designed multiple Lyapunov function's value after switching instants is guaranteed to be smaller than that just before the switching. Moreover, special analytical techniques are utilized to prove that the multiple Lyapunov function is strictly decreasing as the DN evolves. Hence, the finite-time stability theorem in [42] can be easily applied to the issue considered in this paper.

The rest of this technical correspondence is organized as follows. Section II presents model formulation and some preliminary works. Section III develops two synchronization criteria by strict mathematical proofs. In Section IV, numerical simulations are provided to show the effectiveness of our results. Finally, in Section V, conclusions are given.

Notations: The notations used in this paper are quite standard. If not specified, I denotes the nN -dimensional identity matrix. The superscript T stands for transposition, $\mathbb{R}^n$ and $\mathbb{R}^{n \times m}$ denote the n

dimensional Euclidean space and the set of all $n \times m$ real matrices, respectively. In addition, in symmetric block matrices or matrix expressions, we use $*$ as an ellipsis for the terms that are introduced by symmetry. $\text{diag}(\dots)$ stands for a block-diagonal matrix. Matrices, if their dimensions are not explicitly stated, are assumed to be compatible for algebraic operations. $A > 0$ means A is a real symmetric positive definite matrix, $\text{sym}(A) = A + A^T$. $\lambda_{\max}(A)$ ($\lambda_{\min}(A)$) denotes the maximum (minimum) eigenvalue of symmetric matrix A . $\otimes$ is the Kronecker product. $\{t_k, k \in \mathbb{N}_+\}$ stands for a time sequence, where $\mathbb{N}_+ = \{0, 1, 2, \dots\}$. For time-varying function $g(t)$, let $g(t_k^+) = \lim_{t \rightarrow t_k^+} g(t)$, $g(t_k^-) = \lim_{t \rightarrow t_k^-} g(t)$.

II. PRELIMINARIES

Consider the following ordinary differential equation:

$$\dot{x} = h(x), \quad x(0) = x_0 \in \mathbb{R}^d \quad (1)$$

where $h(\cdot) : \mathbb{R}^d \rightarrow \mathbb{R}^d$ is not necessarily continuous. In this technical correspondence, the solutions of (1) are understood in the framework of the following differential inclusion [43]:

$$\dot{x} \in \mathcal{K}[h](x) \triangleq \bigcap_{\delta > 0} \bigcap_{\mu(\Omega)=0} \overline{\text{co}}[h(B(x, \delta) \setminus \Omega)]$$

where $B(x, \delta) = \{y : \|y - x\| \leq \delta\}$, and $\mu(\Omega)$ is the Lebesgue measure of set Ω .

According to [43], the (1) has at least one Filippov solution on $\mathbb{R}^d$, i.e., there exists a measurable function $\gamma \in \mathcal{K}[h](x)$ such that

$$\dot{x} = \gamma. \quad (2)$$

A nonlinear system can be described by

$$\dot{x}(t) = Cx(t) + Af(x(t)) \quad (3)$$

where $x(t) = (x_1(t), \dots, x_n(t))^T \in \mathbb{R}^n$ represents the state vector; $f(x) \in \mathbb{R}^n$ is a continuous function, $C, A \in \mathbb{R}^{n \times n}$ are constant matrices. Given any initial value $x(0) = \varphi \in \mathbb{R}^n$, the trajectory of (3) can be any desired state: equilibrium point, a nontrivial periodic orbit, or even a chaotic orbit.

Consider a controlled network consisting of N coupled systems (3) as follows [5], [15], [16]:

$$\begin{aligned} \dot{x}_i(t) &= Cx_i(t) + Af(x_i(t)) + \sum_{l=1}^N u_{ij} \Phi x_j(t) \\ &+ R_i(t), \quad i = 1, 2, \dots, N \end{aligned} \quad (4)$$

where $x_i(t) = (x_{i1}(t), \dots, x_{in}(t))^T \in \mathbb{R}^n$ represents the state vector of the i th node; $\Phi \in \mathbb{R}^{n \times n}$ is inner coupling matrix between coupled nodes. $U = (u_{ij})_{N \times N}$ is outer-coupling matrix which satisfies the following diffusive condition: $u_{ij} \geq 0$ ($i \neq j$), $u_{ii} = -\sum_{j=1, j \neq i}^N u_{ij}$, ($i, j = 1, 2, \dots, N$). $R_i(t)$ is controller to be designed. The initial value of (4) is $x_i(0) = \varphi_i \in \mathbb{R}^n$, $i = 1, 2, \dots, N$.

Our aim is to design controller $R_i(t)$ such that $x_1(t) = \dots = x_N(t) = x(t)$ in a desired time stated as follows.

Definition 1: The DN (4) is said to be synchronized with (3) in finite time if there exists a constant $0 < \bar{T} < \infty$ [$\bar{T} > 0$ depends on the initial values of (3) and (4)] such that $\lim_{t \rightarrow \bar{T}} \|x_i(t) - x(t)\| = 0$ and $\|x_i(t) - x(t)\| \equiv 0$ for $t > \bar{T}$, $i = 1, 2, \dots, N$, where $\bar{T}$ is called the settling time.

Remark 1: There is essential difference between the finite-time problem in this paper and those considered in [39] and [41]. The finite-time issue in [39] and [41] is to control a system such that the state of the system is finite-time bounded, while the finite-time synchronization in this paper is to control the DN such that the synchronization error approaches to zero in a finite time interval.

Obviously, the techniques in [39] and [41] cannot be applied to the problem stated in Definition 1.

Subtracting (3) from (4) leads to error system

$$\begin{aligned} \dot{e}_i(t) = & Ce_i(t) + Ag(e_i(t)) + \sum_{l=i}^N u_{ij}\Phi e_j(t) \\ & + R_i(t), \quad i = 1, 2, \dots, N \end{aligned} \quad (5)$$

where $e_i(t) = x_i(t) - x(t)$, $g(e_i(t)) = f(x_i(t)) - f(x(t))$.

In view of Definition 1, finite-timely synchronizing (4) with (3) is equivalent to stabilizing the error system (5) to zero solution in finite time.

In order to make full use of interactions between nodes and reduce control cost, intermittent controllers are designed for a part of nodes in (4). Rearrange the order of the nodes and take the first l nodes to be controlled, then design intermittent controllers in the following form:

$$R_i(t) = \begin{cases} -K_i q(e_i(t)) - \eta \text{sign}(q(e_i(t))) \\ \quad t \in [t_{2k}, t_{2k+1}), i = 1, 2, \dots, l \\ -\eta \text{sign}(q(e_i(t))), t \in [t_{2k+1}, t_{2k+2}) \\ \quad i = l+1, l+2, \dots, N \end{cases} \quad (6)$$

where K_i are positive constants to be determined, $\eta > 0$ is a tunable constant, the time sequence $\{t_k, k \in \mathbb{N}_+\}$ satisfies $0 = t_0 < t_1 < \dots < t_k < \dots$, $\lim_{k \rightarrow +\infty} t_k = +\infty$, $k \in \mathbb{N}_+$. $q(e_i(t)) = (q(e_{i1}(t)), \dots, q(e_{in}(t)))^T$, $q(\cdot) : \mathbb{R} \rightarrow \Gamma$ is a quantizer, where $\Gamma = \{\pm w_i : w_i = \rho^i w_0, i = 0, \pm 1, \pm 2, \dots\} \cup \{0\}$ with $w_0 > 0$. $\text{sign}(\cdot)$ is the standard sign function, and $\text{sign}(q(e_i(t))) = (\text{sign}(q(e_{i1}(t))), \dots, \text{sign}(q(e_{in}(t))))^T$. For $\forall v \in \mathbb{R}$, the quantizer $q(v)$ is defined as follows:

$$q(v) = \begin{cases} w_i, & \text{if } \frac{1}{1+\delta} w_i < v \leq \frac{1}{1-\delta} w_i \\ 0, & \text{if } v = 0 \\ -q(-v), & \text{if } v < 0 \end{cases}$$

where $\delta = [(1-\rho)/(1+\rho)]$, $0 < \rho < 1$. According to (1) and (2), there exists a Filippov solution $\Delta \in [-\delta, \delta]$ such that $q(v) = (1+\Delta)v$. It is assumed that there is no jump of the states $x_i(t)$, $i = 1, 2, \dots, N$, at any time instants t_k , $k \in \mathbb{N}_+$. Hence, $e_i(t)$, $i = 1, 2, \dots, N$, are continuous for $t \geq 0$.

Remark 2: The controller (6) is called quantized semi-pinning intermittent controller, which is more practical than the continuously semi-pinning controller in [33] as follows:

$$R_i(t) = \begin{cases} -K_i e_i(t) - \eta \text{sign}(e_i(t)), i = 1, 2, \dots, l \\ -\eta \text{sign}(e_i(t)), i = l+1, l+2, \dots, N. \end{cases}$$

In practice, signal quantization is necessary because the arithmetic in practical control system is finite and capacity of communication channels is limit. Note that there are uncertain differences between $q(e_i(t))$ and $e_i(t)$. By using the special property of sign function, these differences will be overcome in this paper.

Let $\mathbf{g}(e(t)) = (g^T(e_1(t)), \dots, g^T(e_N(t)))^T$, $e(t) = (e_1^T(t), \dots, e_N^T(t))^T$, $\mathbf{C} = I_N \otimes C$, $\mathbf{A} = I_N \otimes A$, $\mathbf{U} = U \otimes \Phi$, $\text{sign}(q(e(t))) = (\text{sign}^T(q(e_1(t))), \dots, \text{sign}^T(q(e_N(t))))^T$, $\mathbf{K} = \text{diag}(K_1, \dots, K_l, 0, \dots, 0) \otimes I_n$. Then the error system (5) with controller (6) can be written in the Kronecker product form [44]

$$\dot{e}(t) = \begin{cases} W(t)e(t) + \mathbf{A}\mathbf{g}(e(t)) - \eta \text{sign}(q(e(t))) \\ \quad t \in [t_{2k}, t_{2k+1}), k \in \mathbb{N}_+ \\ (\mathbf{C} + \mathbf{U})e(t) + \mathbf{A}\mathbf{g}(e(t)) - \eta \text{sign}(q(e(t))) \\ \quad t \in [t_{2k+1}, t_{2k+2}), k \in \mathbb{N}_+ \end{cases} \quad (7)$$

where $\Lambda(t) = \text{diag}(\Lambda_1(t), \Lambda_2(t), \dots, \Lambda_N(t))$, $W(t) = \mathbf{C} + \mathbf{U} - \mathbf{K}(I + \Lambda(t))$, $\Lambda_i(t) = \text{diag}(\Delta_{i1}(t), \Delta_{i2}(t), \dots, \Delta_{in}(t))$, $\Delta_{ij}(t)$ are Filippov solutions with $\Delta_{ij}(t) \in [-\delta, \delta]$, $i = 1, 2, \dots, N$, $j = 1, 2, \dots, n$. Define $F(t) = \Lambda(t)\delta^{-1}$, then

$$F(t)F^T(t) \leq I. \quad (8)$$

The following conditions and lemmas are useful.

(H₁) There exists constant $L > 0$ such that

$$\|f(x) - f(y)\| \leq L\|x - y\|, \quad \forall x, y \in \mathbb{R}^n.$$

(H₂) There exist constants $\tau_{\min} > 0$, $\tau_{\max} > 0$ such that $t_{2k+1} - t_{2k} \geq \tau_{\min}$, $t_{2(k+1)} - t_{2k+1} \leq \tau_{\max}$, $k \in \mathbb{N}_+$.

Remark 3: Condition (H₂) can also be applied to switching systems. Recently, finite-time synchronization of coupled neural networks with arbitrary switching was considered in [37] by designing state feedback controllers. Generally speaking, results in the framework of arbitrary switching are more conservative than those obtained by using dwell time constraint since stability of systems with arbitrary switching require all modes to be stable. Hence, this paper improves the results in [37] to some extent.

Lemma 1 [45]: For given matrices E , F , and G with $F^T F \leq I$ and scalar $\varepsilon > 0$, the following inequality holds:

$$GFE + E^T F^T G^T \leq \varepsilon GG^T + \varepsilon^{-1} E^T E.$$

Lemma 2 [42]: Assume that a continuous, positive-definite function $V(t)$ satisfies the following differential inequality:

$$\dot{V}(t) \leq -\alpha V^\beta(t), \quad \forall t \geq t_0, \quad V(t_0) \geq 0$$

where $\alpha > 0$, $0 < \beta < 1$ are two constants. Then, for any given t_0 , $V(t)$ satisfies the equality $V(t) \equiv 0$, $\forall t \geq T_1$, with T_1 given by

$$T_1 = t_0 + \frac{V^{1-\beta}(t_0)}{\alpha(1-\beta)}.$$

III. MAIN RESULTS

By designing multiple Lyapunov functions and using convex combination method, this section gives sufficient conditions to guarantee the finite-time stabilization of (7). A corollary is derived to show that, under the quantized intermittent semi-pinning controller (6) with $\eta = 0$, i.e., quantized intermittent pinning control, the multiple Lyapunov functions are strict decreasing and asymptotic synchronization is achieved.

For given positive integer D_1 , denote $t_{2k,d_1} = t_{2k} + d_1 \tau_{\min}/D_1$, where $d_1 = 0, 1, 2, \dots, D_1 - 1$. Then $t_{2k,0} = t_{2k}$ and $t_{2k,D_1} = t_{2k} + \tau_{\min}$. Define matrix function

$$P_1(t) = (1 - \alpha_1)P_{1,d_1} + \alpha_1 P_{1,d_1+1} = P_{1,d_1}(\alpha_1), t \in \mathcal{W}_{2k,d_1}$$

where $P_{1,d_1} > 0$, $d_1 = 0, 1, 2, \dots, D_1$, are matrices [46], [47]. $\mathcal{W}_{2k,d_1} = [t_{2k,d_1}, t_{2k,d_1+1})$, $\alpha_1 = (t - t_{2k,d_1})D_1/\tau_{\min}$. It follows that $0 \leq \alpha_1 \leq 1$ and $\dot{\alpha}_1 = D_1/\tau_{\min}$. Moreover, for $t \in [t_{2k}, t_{2k+1})$, $k \in \mathbb{N}_+$, define matrix

$$P_1(t) = \begin{cases} P_{1,d_1}(\alpha_1), t \in \mathcal{W}_{2k,d_1} \\ P_{1,D_1}, t \in [t_{2k,D_1}, t_{2k+1}). \end{cases}$$

Given positive integers D_2 , take $t_{2k+1,d_2} = t_{2k+1} + d_2 \tau_{\max}/D_2$, $d_2 = 0, 1, 2, \dots, D_2 - 1$. For matrices $P_{2,d_2} > 0$, $d_2 = 0, 1, 2, \dots, D_2 - 1$, choose

$$P_2(t) = (1 - \alpha_2)P_{2,d_2} + \alpha_2 P_{2,d_2+1}, t \in \mathcal{W}_{2k+1,d_2}$$

where $\alpha_2 = (t - t_{2k+1,d_2})D_2/\tau_{\max}$ with $\dot{\alpha}_2 = D_2/\tau_{\max}$, $\mathcal{W}_{2k+1,d_2} = [t_{2k+1,d_2}, t_{2k+1,d_2+1})$.

Similarly, for constants $\varepsilon_{i,d_i} > 0$, $\eta_{i,d_i} > 0$, $i = 1, 2$, $d_i = 0, 1, 2, \dots, D_i$, define

$$\varepsilon_1(t) = \begin{cases} \varepsilon_{1,d_1}(\alpha_1), & t \in \mathcal{W}_{2k,d_1} \\ \varepsilon_{1,D_1}, & t \in [t_{2k,D_1}, t_{2k+1}] \end{cases}$$

$$\eta_1(t) = \begin{cases} \eta_{1,d_1}(\alpha_1), & t \in \mathcal{W}_{2k,d_1} \\ \eta_{1,D_1}, & t \in [t_{2k,D_1}, t_{2k+1}] \end{cases}$$

and

$$\varepsilon_2(t) = (1 - \alpha_2)\varepsilon_{2,d_2} + \alpha_2\varepsilon_{2,d_2+1} = \varepsilon_{2,d_2}(\alpha_2)$$

where $\varepsilon_{1,d_1}(\alpha_1) = (1 - \alpha_1)\varepsilon_{1,d_1} + \alpha_1\varepsilon_{1,d_1+1}$, $\eta_{1,d_1}(\alpha_1) = (1 - \alpha_1)\eta_{1,d_1} + \alpha_1\eta_{1,d_1+1}$.

It is obvious that $P_i(t) > 0$, $\varepsilon_i(t) > 0$ ($i = 1, 2$), $\eta_1(t) > 0$, and $\dot{P}_i(t) = \dot{P}_{i,d_i}(\alpha_i) = (D_i/\tau_{\min})(P_{i,d_i+1} - P_{i,d_i})$ for $t \geq 0$. For the sake of simplicity, denote $\mathcal{D}_{1,d_1} = (D_1/\tau_{\min})(P_{1,d_1+1} - P_{1,d_1})$, $\mathcal{D}_{2,d_2} = (D_2/\tau_{\max})(P_{2,d_2+1} - P_{2,d_2})$.

Our main results are stated as follows.

Theorem 1: Assume that the conditions (H₁) and (H₂) are satisfied and $\eta > 0$. Then, for given control gain $\mathbf{K}$ and positive integers D_1, D_2 , the network (4) with controller (6) can be synchronized with (3) in finite time if there exist positive definite diagonal matrices P_{i,d_i} , and scalars $\varepsilon_{i,d_i} > 0$, $\eta_{1,d_1} > 0$, $d_i = 0, 1, \dots, D_i - 1$, $i = 1, 2$, such that

$$\Sigma_{d_1} = \begin{pmatrix} \Sigma_{1,d_1} & P_{1,d_1}\mathbf{A} & P_{1,d_1}\mathbf{K}\delta \\ * & -\varepsilon_{1,d_1}I & 0 \\ * & * & -\eta_{1,d_1}I \end{pmatrix} < 0 \quad (9)$$

$$\Sigma_{d_1+1} = \begin{pmatrix} \Sigma_{1,d_1+1} & P_{1,d_1+1}\mathbf{A} & P_{1,d_1+1}\mathbf{K}\delta \\ * & -\varepsilon_{1,d_1+1}I & 0 \\ * & * & -\eta_{1,d_1+1}I \end{pmatrix} < 0 \quad (10)$$

$$\Sigma_{D_1} = \begin{pmatrix} \Sigma_{1,D_1} & P_{1,D_1}\mathbf{A} & P_{1,D_1}\mathbf{K}\delta \\ * & -\varepsilon_{1,D_1}I & 0 \\ * & * & -\eta_{1,D_1}I \end{pmatrix} < 0 \quad (11)$$

$$\Sigma_{d_2} = \begin{pmatrix} \widehat{\Sigma}_{2,d_2} & P_{2,d_2}\mathbf{A} \\ * & -\varepsilon_{2,d_2}I \end{pmatrix} < 0 \quad (12)$$

$$\Sigma_{d_2+1} = \begin{pmatrix} \widehat{\Sigma}_{2,d_2+1} & P_{2,d_2+1}\mathbf{A} \\ * & -\varepsilon_{2,d_2+1}I \end{pmatrix} < 0 \quad (13)$$

and

$$P_{1,0} < P_{2,D_2}, \quad P_{2,0} < P_{1,D_1} \quad (14)$$

where $\Sigma_{1,d_1} = (\varepsilon_{1,d_1}L^2 + \eta_{1,d_1})I + \mathcal{D}_{1,d_1} + \text{sym}(P_{1,d_1}(\mathbf{C} + \mathbf{U} - \mathbf{K}))$, $\Sigma_{1,d_1+1} = (\varepsilon_{1,d_1+1}L^2 + \eta_{1,d_1+1})I + \mathcal{D}_{1,d_1} + \text{sym}(P_{1,d_1+1}(\mathbf{C} + \mathbf{U} - \mathbf{K}))$, $\Sigma_{1,D_1} = (\varepsilon_{1,D_1}L^2 + \eta_{1,D_1})I + \text{sym}(P_{1,D_1}(\mathbf{C} + \mathbf{U} - \mathbf{K}))$, $\widehat{\Sigma}_{2,d_2} = \varepsilon_{2,d_2}L^2I + \mathcal{D}_{2,d_2} + \text{sym}(P_{2,d_2}(\mathbf{C} + \mathbf{U}))$, $\widehat{\Sigma}_{2,d_2+1} = \varepsilon_{2,d_2+1}L^2I + \mathcal{D}_{2,d_2} + \text{sym}(P_{2,d_2+1}(\mathbf{C} + \mathbf{U}))$. Moreover, the settling time is estimated as $\bar{T} = ((e^T(0)P_{1,0}e(0))^{0.5}/\eta\gamma)$, where $\gamma = \min\{\underline{\lambda}_q\rho_q, q = 1, 2\}$, $\underline{\lambda}_q = \min\{\lambda_{\min}(P_{q,d_q}), d_q = 0, 1, \dots, D_q\}$, $\bar{\lambda}_q = \max\{\lambda_{\max}(P_{q,d_q}), d_q = 0, 1, \dots, D_q\}$, $\rho_q = (1/\sqrt{\bar{\lambda}_q})$, $q = 1, 2$.

Proof: For $\forall k \in \mathbb{N}_+$, consider multiple Lyapunov function

$$V(t) = \begin{cases} e^T(t)P_1(t)e(t), & t_{2k} \leq t < t_{2k+1} \\ e^T(t)P_2(t)e(t), & t_{2k+1} \leq t < t_{2k+2} \end{cases} \quad (15)$$

It is derived from (7) that, for $t \in [t_{2k}, t_{2k} + \tau_{\min}]$, $k \in \mathbb{N}_+$

$$\dot{V}(t) = e^T(t)\dot{P}_1(t)e(t) + 2e^T(t)P_1(t)[W(t)e(t) + \mathbf{A}g(e(t)) - \eta\text{sign}(q(e(t)))]. \quad (16)$$

In view of (H₁), it holds that, for any $\varepsilon_1(t) > 0$

$$\varepsilon_1(t)L^2e^T(t)e(t) - \varepsilon_1(t)\mathbf{g}^T(e(t))\mathbf{g}(e(t)) \geq 0. \quad (17)$$

Since $\delta \in (0, 1)$, one has $\text{sign}(q(e(t))) = \text{sign}(e(t))$. Then

$$\begin{aligned} -2\eta e^T(t)P_1(t)\text{sign}(q(e(t))) &\leq -2\eta\underline{\lambda}_1 \sum_{i=1}^N \sum_{j=1}^n |e_{ij}(t)| \\ &= -2\underline{\lambda}_1 \|e(t)\|_1 \\ &\leq -2\eta\underline{\lambda}_1 \rho_1 V^{\frac{1}{2}}(t) \end{aligned} \quad (18)$$

where $e^T(t)P_1(t)e(t) \leq \bar{\lambda}_1 \|e(t)\|_2^2 \leq \bar{\lambda}_1 \|e(t)\|_1^2$ has been used.

Combining (16)–(18) gives that, for $t \in [t_{2k}, t_{2k} + \tau_{\min})$

$$\dot{V}(t) \leq \xi^T(t)\Sigma(t)\xi(t) - 2\eta\underline{\lambda}_1 \rho_1 V^{\frac{1}{2}}(t) \quad (19)$$

where $\xi(t) = (e^T(t), \mathbf{g}^T(e(t)))^T$, $\widehat{\Sigma}_{11}(t) = \varepsilon_1(t)L^2I + (D_1/\tau_{\min})(P_{1,d_1+1} - P_{1,d_1}) + P_1(t)W(t) + W^T(t)P_1(t)$, and $\Sigma(t) = \begin{pmatrix} \widehat{\Sigma}_{11}(t) & P_1(t)\mathbf{A} \\ * & -\varepsilon_1(t)I \end{pmatrix}$.

By the Schur Complement [48] and conditions (9) and (10), one has

$$\begin{aligned} \Sigma(t) &= \begin{pmatrix} \widehat{\Sigma}_{11}(t) & P_1(t)\mathbf{A} \\ * & -\varepsilon_1(t)I \end{pmatrix} \\ &\quad + M(t)F(t)E^T + EF(t)M^T(t) \\ &\leq \begin{pmatrix} \Sigma_{11}(t) & P_1(t)\mathbf{A} & P_1(t)\mathbf{K}\delta \\ * & -\varepsilon_1(t)I & 0 \\ * & * & -\eta_1(t)I \end{pmatrix} \\ &= (1 - \alpha_1)\Sigma_{d_1} + \alpha_1\Sigma_{d_1+1} < 0 \end{aligned} \quad (20)$$

where convex combination technique has been used, $M(t) = (-\delta(P_1(t)\mathbf{K})^T, 0)^T$, $E = (I, 0)^T$, $\widehat{\Sigma}_{11}(t) = \varepsilon_1(t)L^2I + (D_1/\tau_{\min})(P_{1,d_1+1} - P_{1,d_1}) + P_1(t)(\mathbf{C} + \mathbf{U} - \mathbf{K}) + (\mathbf{C} + \mathbf{U} - \mathbf{K})^T P_1(t)$, $\Sigma_{11}(t) = \widehat{\Sigma}_{11}(t) + \eta_1(t)I$.

It is found from (19) and (20) that, for $t \in [t_{2k}, t_{2k} + \tau_{\min})$

$$\dot{V}(t) \leq -2\eta\underline{\lambda}_1 \rho_1 V^{\frac{1}{2}}(t), \quad t \in [t_{2k}, t_{2k} + \tau_{\min}). \quad (21)$$

By similar discussions above, inequality (11) guarantees that

$$\dot{V}(t) \leq -2\eta\underline{\lambda}_1 \rho_1 V^{\frac{1}{2}}(t), \quad t \in [t_{2k,D_1}, t_{2k+1}). \quad (22)$$

Therefore, it is natural to derive from (21) and (22) that

$$\dot{V}(t) \leq -2\eta\underline{\lambda}_1 \rho_1 V^{\frac{1}{2}}(t), \quad t \in [t_{2k}, t_{2k+1}). \quad (23)$$

Similarly, when $t \in [t_{2k+1}, t_{2k+2})$, $\dot{V}(t) \leq -2\eta\underline{\lambda}_2 \rho_2 V^{(1/2)}(t)$ can also be guaranteed by inequalities (12) and (13).

It is time to consider the instants $t = t_{2k}$ and $t = t_{2k+1}$, $k \in \mathbb{N}_+$. When $t = t_{2k}$, it can be derived that $P_1(t_{2k}) = P_{1,0}$, $P_2(t_{2k}) = P_{2,D_2}$. When $t = t_{2k+1}$, it can be derived that $P_1(t_{2k+1}) = P_{1,D_1}$, $P_2(t_{2k+1}) = P_{1,0}$. Since $e(t)$ is continuous at t_{2k} and t_{2k+1} , condition (14) guarantees that $V(t_{2k}^-) > V(t_{2k}^+)$ and $V(t_{2k+1}^-) > V(t_{2k+1}^+)$, $k \in \mathbb{N}_+$. Keeping in mind that $\dot{V}(t) \leq -2\eta\gamma V^{(1/2)}(t)$ for $t \neq t_k$, it is derived from Lemma 2 that $V(t) \equiv 0$ for $t \geq \bar{T}$. Therefore, $\lim_{t \rightarrow \bar{T}} \|e_i(t)\| = 0$ and $\|e_i(t)\| \equiv 0$ for $t > \bar{T}$, $i = 1, 2, \dots, N$. This completes the proof. ■

Note that, when the values of D_1 and D_2 is large, Theorem 1 has to solve a large number of LMIs. In the case that the network has a large number of nodes, solving these LMIs is time-consuming. Hence, it is difficult to apply the synchronization criteria in Theorem 1 to DNS with large number of nodes.

If only asymptotic synchronization is considered, then η can be taken as zero. In this case, the semi-pinning controller (6) becomes a quantized intermittent pinning controller. It is easy to derive the following result from Theorem 1, which still improve the results obtained in [11]–[13] to some extent, since quantizer was not considered in [11]–[13]. Moreover, the positive definite diagonal matrices $P_{i,d_i} > 0$ in Theorem 1 can be relaxed to positive definite matrices.

Corollary 1: Assume that conditions (H_1) and (H_2) are satisfied and control gain $\mathbf{K}$ and positive integers D_1, D_2 are given. Then, under controller (6) with $\eta = 0$, the network (4) can be asymptotically synchronized with (3) if there exist positive definite diagonal matrices $P_{i,d_i} > 0$, and scalars $\varepsilon_{i,d_i} > 0, \eta_{1,d_1} > 0, d_i = 0, 1, \dots, D_i - 1, i = 1, 2$, such that the inequalities (9)–(14) hold.

Remark 4: From the proof of Theorem 1 one can see that, with the aid of convex combination technique, the designed multiple Lyapunov function (15) is strictly decreasing before the realization of the synchronization, which is completely different from most of existing literatures concerning synchronization of DNs via pinning control where the designed Lyapunov functions increase exponentially when there is no control and decrease exponentially when the control works. Moreover, the analytical techniques are different from those used in [11]–[13] and [21]–[25], where asymptotic synchronization of DNs has been investigated by directly using state feedback control. On the other hand, compared with the methods utilized in [11]–[13] and [21]–[25], the multiple Lyapunov function and convex combination techniques can reduce the conservativeness of obtained result because it is not necessary to estimate the largest increasing rate and the least decreasing rate of the Lyapunov functions in uncontrolled and controlled intervals, respectively.

Remark 5: For given D_1, D_2 , and η , the settling time in Theorem 1 depends on the values of $P_{1,0}$ and $\gamma = \min\{\lambda_q \rho_q, q = 1, 2\}$. It is found that larger value of γ leads to more accurate estimation of the settling time. Hence, an optimal algorithm is given as follows.

- 1) For given $0 < z < 1$, suppose there exist positive constants $\varrho_{i,d_i}, d_i = 0, 1, \dots, D_i, i = 1, 2$, such that

$$P_{i,d_i} - \varrho_{i,d_i} I < 0 \text{ and } z\varrho_{i,d_i} I - P_{i,d_i} < 0. \quad (24)$$

- 2) $z_0 = \max_{0 < z < 1} \{z : (9)–(14) \text{ and } (24) \text{ hold}\}$.
- 3) Use the $P_{1,0}$ obtained by solving step 2) to compute the settling time according to the formula $\bar{T} = ((e^T(0)P_{1,0}e(0))^{0.5}/\eta\gamma)$.

IV. NUMERICAL EXAMPLE

In this section, one numerical example is given to illustrate Theorem 1.

The Chua's circuit is described as follows [49]:

$$\dot{x}(t) = Cx(t) + Af(x(t)) \quad (25)$$

where $x(t) = (x_1(t), x_2(t), x_3(t))^T$, $A = \text{diag}(54/7, 0, 0)$, $f(x(t)) = 0.5(|x_1(t) + 1| - |x_1(t) - 1|, 0, 0)^T$

$$C = \begin{pmatrix} -\frac{19}{7} & 9 & 0 \\ 1 & -1 & 1 \\ 0 & -14.28 & 0 \end{pmatrix}.$$

The trajectory of (25) with initial condition $x(0) = (0.65, 0.2, 0.8)^T$ is shown in Fig. 1.

Consider a DN as follows:

$$\dot{x}_i(t) = Cx_i(t) + Af(x_i(t)) + \sum_{l=1}^5 u_{ij}\Phi x_j(t) + R_i(t) \quad (26)$$

$$i = 1, 2, \dots, 5$$

where $x_i(t) = (x_{i1}(t), x_{i2}(t), x_{i3}(t))^T$, $\Phi = \text{diag}(1, 1, 1)$

$$U = 7 \begin{pmatrix} -2 & 0 & 1 & 0 & 1 \\ 1 & -3 & 1 & 0 & 1 \\ 1 & 1 & -4 & 1 & 1 \\ 1 & 1 & 0 & -4 & 2 \\ 1 & 2 & 1 & 1 & -5 \end{pmatrix}.$$

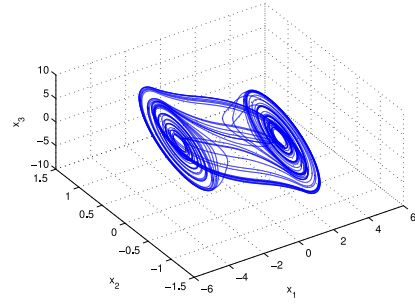


Fig. 1. Trajectory of isolated Chua's circuit with initial value $x(0) = (0.65, 0.2, 0.8)^T$.

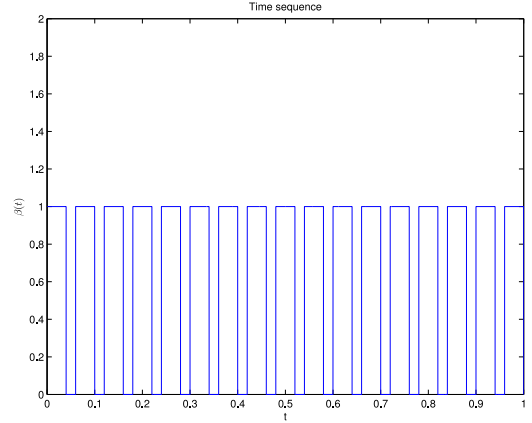


Fig. 2. Control intervals with $\tau_{\min} = 0.04$, $\tau_{\max} = 0.02$. The controller (6) is active when $\beta = 1$ and it is inactive when $\beta = 0$.

By simple computation, the condition (H_1) is satisfied with $L = 1$. The time sequence $\{t_k, k \in \mathbb{N}_+\}$ is assumed to satisfy that $t_{2k+1} - t_{2k} = \tau_{\min} = 0.04$ and $t_{2(k+1)} - t_{2k+1} = \tau_{\max} = 0.02, k \in \mathbb{N}_+$. Then the controller (6) is active when $t \in [k0.06, k0.06 + 0.04]$ while it is inactive when $t \in [k0.06 + 0.04, (k+1)0.06]$. The time sequence is shown in Fig. 2.

Take the quantizer density as $\rho = 0.8$, $D_1 = 2$, and $D_2 = 3$, the control gains in (6) are $K_1 = 25$, $K_2 = K_3 = 20$, $K_4 = K_5 = 0$, and $\eta = 1$.

By Remark 5, solving the LMIs (9)–(14) and (24) in MATLAB derives that $z_0 = 0.0553$ and feasible solutions with $\mathbf{P}_{1,0} = 10^{-9} \text{diag}(2.01, 13.79, 1.41, 1.88, 13.78, 1.35, 1.36, 9.78, 0.96, 1.27, 9.18, 1.08, 1.82, 14.74, 1.32)$, $\lambda_1 \rho_1 = 7.9359 \times 10^{-6}$, and $\lambda_2 \rho_2 = 7.9512 \times 10^{-6}$. Hence, $\gamma = 7.9359 \times 10^{-6}$.

The initial value of (26) is randomly chosen as $\varphi = (-2.8649, -0.6259, 3.3259, 2.3377, 3.6759, 1.2459, -3.7143, 2.793, 3.4719, 1.4299, 2.0619, 1.9451, -0.8622, 1.2438, -2.6305)^T$. According to Theorem 1, the DN (26) with quantized controller (6) can be synchronized onto (25) in settling time $\bar{T} = 81.8377$.

Taking step size 0.001, we have the trajectories $q(e_i(t))$ and $e_i(t)$ $i = 1, 2, 3, 4, 5$ shown in Fig. 3, which demonstrates that the synchronization is realized in the $\bar{T}$.

Remark 6: If we take $D_1 = D_2 = 0$ and $P_{1,0} = P_{2,0} = I$, then the multiple Lyapunov function (15) in Theorem 1 becomes $V_i(t) = e^T(t)e(t)$ which was utilized in [11]–[13] and [21]–[23] for state feedback intermittent pinning synchronization. However, in this example, the LMIs (9)–(14) have no feasible solution when $D_1 = D_2 = 0$. Therefore, results in [11]–[13] and [21]–[23] are not applicable to this example. Actually, the LMIs (9)–(14) have no feasible solution when $D_1 \leq 1$ and $D_2 \leq 2$. Moreover, by numerical

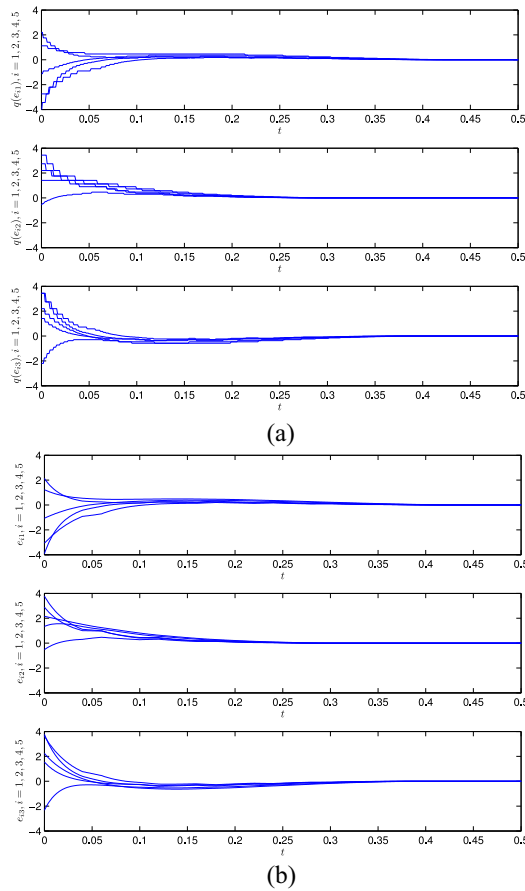


Fig. 3. Trajectories $q(e_i(t))$ (a) and $e_i(t)$ (b) between the network (26) and isolated system (25), where the first three subsystems are controlled.

computation, the larger the values of D_1 and D_2 are, the less the conservativeness of Theorem 1 will be.

V. CONCLUSION

Finite-time synchronization of DNs with aperiodically intermittent quantized semi-pinning controller has been investigated in this technical correspondence. By designing multiple Lyapunov functions and using convex combination technique, sufficient conditions in terms of LMIs have been derived. Compared with existing pinning control, the quantized controller needs a limited information. Moreover, the designed multiple Lyapunov function's value is strictly decreasing, whether the controller works or not. These special analytical techniques make it easy to use classical finite-time stability theorem. Numerical example and simulations demonstrated the effectiveness of the theoretical results.

The sign function in the controller (6) will induce chattering phenomenon, which brings bad effect to systems in practice. How to design quantized controllers without using the sign function to achieve finite-time synchronization is our next research topic, which is also challenging.

REFERENCES

- [1] T.-L. Liao and N.-S. Huang, "An observer-based approach for chaotic synchronization with applications to secure communications," *IEEE Trans. Circuits Syst. I, Fundam. Theory Appl.*, vol. 46, no. 9, pp. 1144–1150, Sep. 1999.
- [2] M. Abeles, Y. Prut, H. Bergman, and E. Vaadia, "Synchronization in neuronal transmission and its importance for information processing," *Progr. Brain Res.*, vol. 102, pp. 395–404, Aug. 1994.
- [3] L. M. Pecora and T. L. Carroll, "Synchronization in chaotic systems," *Phys. Rev. Lett.*, vol. 64, no. 8, pp. 821–824, Feb. 1990.
- [4] X. Yang and D. W. C. Ho, "Synchronization of delayed memristive neural networks: Robust analysis approach," *IEEE Trans. Cybern.*, vol. 46, no. 12, pp. 3377–3387, Dec. 2016.
- [5] T. Chen, X. Liu, and W. Lu, "Pinning complex networks by a single controller," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 54, no. 6, pp. 1317–1326, Jun. 2007.
- [6] J. Zhou, J.-A. Lu, and J. Lü, "Pinning adaptive synchronization of a general complex dynamical network," *Automatica*, vol. 44, no. 4, pp. 996–1003, 2008.
- [7] Q. Song, F. Liu, J. Cao, and W. Yu, "M-matrix strategies for pinning-controlled leader-following consensus in multiagent systems with nonlinear dynamics," *IEEE Trans. Cybern.*, vol. 43, no. 6, pp. 1688–1697, Dec. 2013.
- [8] J. Wang, J. Feng, C. Xu, Y. Zhao, and J. Feng, "Pinning synchronization of nonlinearly coupled complex networks with time-varying delays using M-matrix strategies," *Neurocomputing*, vol. 177, pp. 89–97, Feb. 2016.
- [9] J. Lu, D. W. C. Ho, and J. Cao, "A unified synchronization criterion for impulsive dynamical networks," *Automatica*, vol. 46, no. 7, pp. 1215–1221, 2010.
- [10] X. Yang, J. Cao, and Z. Yang, "Synchronization of coupled reaction-diffusion neural networks with time-varying delays via pinning-impulsive controller," *SIAM J. Control Optim.*, vol. 51, no. 5, pp. 3486–3510, 2013.
- [11] X. Yang and J. Cao, "Stochastic synchronization of coupled neural networks with intermittent control," *Phys. Lett. A*, vol. 373, no. 36, pp. 3259–3272, 2009.
- [12] X. Liu and T. Chen, "Synchronization of complex networks via aperiodically intermittent pinning control," *IEEE Trans. Autom. Control*, vol. 60, no. 12, pp. 3316–3321, Dec. 2015.
- [13] J. Feng, P. Yang, and Y. Zhao, "Cluster synchronization for nonlinearly time-varying delayed coupling complex networks with stochastic perturbation via periodically intermittent pinning control," *Appl. Math. Comput.*, vol. 291, pp. 52–68, Dec. 2016.
- [14] A. Khan and M. Shahzad, "Synchronization of circular restricted three body problem with Lorenz hyper chaotic system using a robust adaptive sliding mode controller," *Complexity*, vol. 18, no. 6, pp. 58–64, 2013.
- [15] J. Lu, J. Zhong, C. Huang, and J. Cao, "On pinning controllability of Boolean control networks," *IEEE Trans. Autom. Control*, vol. 61, no. 6, pp. 1658–1663, Jun. 2016.
- [16] X. Yang and J. Cao, "Adaptive pinning synchronization of coupled neural networks with mixed delays and vector-form stochastic perturbations," *Acta Math. Sci.*, vol. 32, no. 3, pp. 955–977, 2012.
- [17] C. Hu, J. Yu, H. Jiang, and Z. Teng, "Pinning synchronization of weighted complex networks with variable delays and adaptive coupling weights," *Nonlin. Dyn.*, vol. 67, no. 2, pp. 1373–1385, 2012.
- [18] Q. Song and J. Cao, "On pinning synchronization of directed and undirected complex dynamical networks," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 57, no. 3, pp. 672–680, Mar. 2010.
- [19] X. Liu and T. Chen, "Finite-time and fixed-time cluster synchronization with or without pinning control," *IEEE Trans. Cybern.*, to be published, doi: 10.1109/TCYB.2016.2630703.
- [20] J. Wang, J. Feng, C. Xu, and Y. Zhao, "Exponential synchronization of stochastic perturbed complex networks with time-varying delays via periodically intermittent pinning," *Commun. Nonlin. Sci. Numer. Simulat.*, vol. 18, no. 11, pp. 3146–3157, 2013.
- [21] W. Xia and J. Cao, "Pinning synchronization of delayed dynamical networks via periodically intermittent control," *Chaos*, vol. 19, no. 1, 2009, Art. no. 013120.
- [22] S. Cai, J. Hao, Q. He, and Z. Liu, "Exponential synchronization of complex delayed dynamical networks via pinning periodically intermittent control," *Phys. Lett. A*, vol. 375, no. 19, pp. 1965–1971, 2011.
- [23] X. Liu and T. Chen, "Synchronization of nonlinear coupled networks via aperiodically intermittent pinning control," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 26, no. 1, pp. 113–126, Jan. 2015.
- [24] W. Lu, Y. Han, and T. Chen, "Pinning networks of coupled dynamical systems with Markovian switching couplings and event-triggered diffusions," *J. Frankl. Inst.*, vol. 352, no. 9, pp. 3526–3545, 2015.
- [25] B. Zhou, X. Liao, T. Huang, and G. Chen, "Pinning exponential synchronization of complex networks via event-triggered communication with combinational measurements," *Neurocomputing*, vol. 157, pp. 199–207, Jun. 2015.
- [26] R. W. Brockett and D. Liberzon, "Quantized feedback stabilization of linear systems," *IEEE Trans. Autom. Control*, vol. 45, no. 7, pp. 1279–1289, Jul. 2000.

- [27] F. Ceragioli, C. D. Persis, and P. Frasca, "Discontinuities and hysteresis in quantized average consensus," *Automatica*, vol. 47, no. 9, pp. 1916–1928, 2011.
- [28] S. Liu, T. Li, L. Xie, M. Fu, and J.-F. Zhang, "Continuous-time and sampled-data-based average consensus with logarithmic quantizers," *Automatica*, vol. 49, no. 11, pp. 3329–3336, 2013.
- [29] L. Li, D. W. C. Ho, and J. Lu, "A unified approach to practical consensus with quantized data and time delay," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 60, no. 10, pp. 2668–2678, Oct. 2013.
- [30] W. Liu, Z. Wang, and M. Ni, "Quantized feedback stabilization for a class of linear systems with nonlinear disturbances," *Nonlin. Anal. Hybrid Syst.*, vol. 8, pp. 48–56, May 2013.
- [31] G. Song, T. Li, Y. Li, and J. Lu, "Quantized output feedback stabilization for nonlinear discrete-time systems subject to saturating actuator," *Nonlin. Dyn.*, vol. 83, nos. 1–2, pp. 305–317, 2016.
- [32] Z. Liu, F. Wang, Y. Zhang, and C. L. P. Chen, "Fuzzy adaptive quantized control for a class of stochastic nonlinear uncertain systems," *IEEE Trans. Cybern.*, vol. 46, no. 2, pp. 524–534, Feb. 2016.
- [33] X. Yang, D. W. C. Ho, J. Lu, and Q. Song, "Finite-time cluster synchronization of T–S fuzzy complex networks with discontinuous subsystems and random coupling delays," *IEEE Trans. Fuzzy Syst.*, vol. 23, no. 6, pp. 2302–2316, Dec. 2015.
- [34] X. Yang and J. Lu, "Finite-time synchronization of coupled networks with Markovian topology and impulsive effects," *IEEE Trans. Autom. Control*, vol. 61, no. 8, pp. 2256–2261, Aug. 2016.
- [35] H. Du, Y. He, and Y. Cheng, "Finite-time synchronization of a class of second-order nonlinear multi-agent systems using output feedback control," *IEEE Trans. Circuits Syst. I, Reg. Papers*, vol. 61, no. 6, pp. 1778–1788, Jun. 2014.
- [36] M. Ghasemi and S. G. Nersisov, "Finite-time coordination in multiagent systems using sliding mode control approach," *Automatica*, vol. 50, no. 4, pp. 1209–1216, 2014.
- [37] X. Liu, J. Cao, W. Yu, and Q. Song, "Nonsmooth finite-time synchronization of switched coupled neural networks," *IEEE Trans. Cybern.*, vol. 46, no. 10, pp. 2360–2371, Oct. 2016.
- [38] Z. Xiang, C. Qiao, and M. S. Mahmoud, "Robust H_∞ filtering for switched stochastic systems under asynchronous switching," *J. Frankl. Inst.*, vol. 349, no. 3, pp. 1213–1230, 2012.
- [39] G. Zong, R. Wang, W. X. Zheng, and L. Hou, "Finite-time stabilization for a class of switched time-delay systems under asynchronous switching," *Appl. Math. Comput.*, vol. 219, no. 11, pp. 5757–5771, 2013.
- [40] L. Zhang and P. Shi, "Stability, l_2 -gain and asynchronous H_∞ control of discrete-time switched systems with average dwell time," *IEEE Trans. Autom. Control*, vol. 54, no. 9, pp. 2192–2199, Sep. 2009.
- [41] Y. Wu, J. Cao, Q. Li, A. Alsaedi, and F. E. Alsaadi, "Finite-time synchronization of uncertain coupled switched neural networks under asynchronous switching," *Neural Netw.*, vol. 85, pp. 128–139, Jan. 2017.
- [42] Y. Tang, "Terminal sliding mode control for rigid robots," *Automatica*, vol. 34, no. 1, pp. 51–56, 1998.
- [43] A. Filippov, *Differential Equations With Discontinuous Righthand Sides*. Dordrecht, The Netherlands: Kluwer Acad., 1988.
- [44] R. Horn and C. Johnson, *Matrix Analysis*. Cambridge, U.K.: Cambridge Univ. Press, 1990.
- [45] Y. Wang, L. Xie, and C. E. D. Souza, "Robust control of a class of uncertain nonlinear systems," *Syst. Control Lett.*, vol. 19, no. 2, pp. 139–149, 1992.
- [46] L. I. Allerhand and U. Shaked, "Robust stability and stabilization of linear switched systems with dwell time," *IEEE Trans. Autom. Control*, vol. 56, no. 2, pp. 381–386, Feb. 2011.
- [47] W. Xiang, "On equivalence of two stability criteria for continuous-time switched systems with dwell time constraint," *Automatica*, vol. 54, pp. 36–40, Apr. 2015.
- [48] S. Boyd, L. El Ghaoui, E. Feron, and V. Balakrishnan, *Linear Matrix Inequalities in System and Control Theory*. Philadelphia, PA, USA: SIAM, 1994.
- [49] T. Matsumoto, L. Chua, and M. Komuro, "The double scroll," *IEEE Trans. Circuits Syst.*, vol. CS-32, no. 8, pp. 797–818, Aug. 1985.
- [50] J. Lu, D. W. C. Ho, and L. Wu, "Exponential stabilization of switched stochastic dynamical networks," *Nonlinearity*, vol. 22, no. 4, p. 889, 2009.
- [51] Z. Wang, D. W. C. Ho, Y. Liu, and X. Liu, "Robust H_∞ control for a class of nonlinear discrete time-delay stochastic systems with missing measurements," *Automatica*, vol. 45, no. 3, pp. 684–691, 2009.